

Healthcare Revenue Cycle, Utilization, and Patient Experience Dashboard (AWS + Power BI)

A solo project where I built an AWS-backed analytics pipeline and an interactive Power BI report for healthcare performance KPIs.

Objective:

- Build a cloud analytics workflow from raw files to curated reporting tables.
- Create an AWS data lake flow: S3 storage, Glue Data Catalog, Athena SQL, and Power BI.
- Model curated fact and dimension tables for consistent KPI reporting.
- Design a 3-page Power BI report with filters, drill-ready visuals, and KPI cards.

Tools used for this project:

- Amazon S3 (raw, curated, and Athena query results folders)
- AWS Glue Data Catalog and Crawler (metadata and table discovery)
- Amazon Athena (SQL queries and curated views)
- Power BI Desktop (data model, DAX measures, and report design)
- Jira (task tracking and delivery workflow)
- Git and GitHub (version control and portfolio publishing)

Questions explored throughout this project:

1. How do total charges, allowed amounts, and paid amounts compare across the year?
2. What is the denial rate overall, and which facilities drive denials?
3. How does net paid trend by month, and where are the peak and dip periods?
4. Which payer groups contribute the most to total paid?
5. How do encounter volumes change by month and by encounter type?
6. What share of encounters are inpatient and ED, and how does that shift by facility?
7. What is the 30-day readmission rate, and which facilities show higher rates?
8. How does average length of stay vary by facility and by time?
9. What is the patient experience weighted top box score overall and by facility?
10. How do patient experience domains trend over time, and which domains need focus?

Data Attributes:

Source

De-identified, practice healthcare claims, encounters, and patient experience survey data used for analytics work.

Main fields and meaning

- Fact tables: claim lines, claim headers, encounters, patient experience

- Dimensions: date, facility, payer, patient, provider
- Key metrics: charges, allowed, paid, denials, encounter counts, length of stay, readmissions, survey top box percent, respondents

Limitations

- Sample dataset for portfolio work. Results reflect the dataset scope and definitions.
- Patient experience measures depend on respondent volume, so small slices show more volatility.
- Financial fields reflect claim activity in the dataset, not full cost accounting.

Folder Directory

Dashboard

- HealthcareAWS_PowerBI.pbix (Power BI report file)

SQL

- athena/ (Athena SQL used to validate data and build curated outputs)
- ddl/ (Glue table definitions or Athena CREATE statements if used)
- checks/ (row counts, null checks, duplicate checks)

AWS

- s3_layout.md (raw, curated, and athena-results folder plan)
- glue_crawler_config.md (crawler target, database, and table prefix notes)
- athena_settings.md (query result location and workgroup settings)

Documents

- project_summary.docx (this document)
- architecture.png (simple pipeline diagram)
- data_dictionary.md (table and column descriptions)

Images

- report_screenshots/ (screenshots for README and LinkedIn)
- aws_console/ (S3, Glue, Athena screenshots)
- jira/ (board and task list screenshots)

Project Management

- jira_export.csv (optional export for issues and status)

Key Findings

Revenue Cycle Overview

- Total charges across the selected period were \$11.00M.
- Total allowed was \$5.47M, which is 49.7% of charges (\$5.53M in contractual adjustments).
- Total paid was \$3.85M, which is 70.4% of allowed (\$1.62M allowed-but-unpaid).
- Denial rate was 10.16%, or about \$1.12M of charges denied.
- Payer mix was concentrated. Medicare led at \$1.09M paid, about 28.4% of all payments.
- The top 3 payers (Medicare, Commercial A, Medicaid) totaled about \$2.17M, roughly 56.3% of total paid.
- Denial rates varied by facility from 10.77% to 13.71%, a 2.94 percentage-point spread.

Utilization and Quality

- Total encounters across the selected period were 12.0K.
- Inpatient encounters were 1.65K.
- ED encounters were 2.13K.
- 30-day readmission rate was 10.29%.
- Encounter mix was led by Outpatient at about 54.7% (~7K).
- Inpatient and ED each contributed about 17.8% (~2K each).
- Observation contributed about 13.8% (~2K).

Patient Experience and Quality

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Dashboard overview

The Power BI report includes three pages:

- Revenue Cycle Overview: KPI cards, net paid trend, paid by payer, and denial rate by facility.
- Utilization and Quality: encounter volume trends, encounter mix, readmission rate, and length of stay views.
- Patient Experience and Quality: top box KPI, domain trend panels, and facility-by-domain comparison heatmap.

Skills Demonstrated

- Cloud data organization in S3 with clear raw and curated zones
- Metadata management with Glue Data Catalog and crawlers

- SQL analysis and validation in Athena
- Dimensional modeling with fact and dimension tables
- DAX measures for KPIs and time-based analysis
- Dashboard layout, KPI card design, slicers, and navigation
- Project tracking with Jira and documented deliverables
- Portfolio publishing with GitHub-ready structure

Business Impact and next steps

Business Impact

- Faster visibility into denials, payer performance, and month-to-month payment movement.
- Clear comparisons across facilities for utilization, readmissions, and patient experience domains.
- Single report layout that supports review meetings and follow-up action planning.

Next steps

- Publish the report to Power BI Service and set a refresh schedule against Athena.
- Add row-level security for facility-level access control.
- Add drill-through pages for facility detail, payer detail, and denial reason breakdown.
- Document a full data dictionary and add a simple architecture diagram to the GitHub README.